Assel Kembay

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EDUCATION

University of California, Santa Cruz — **Santa Cruz, CA, USA**
Ph.D. in Electrical and Computer Engineering | GPA: 3.95/4.00 | *Advisor: Prof. Jason Eshraghian* — Jun. 2028 (expected)
Area of Study: energy-efficient AI systems, Brain-inspired AI/ML

Korea University of Science and Technology — **Seoul, South Korea**
M.S. in AI - Robotics | GPA: 4.43/4.50
Thesis: "Inversion of Spiking Neural Networks & its application to Knowledge Distillation"

C-DAC's Advanced Computing Training School — **Pune, India**
Advanced Computing (Exchange student)

L.N. Gumilyov Eurasian National University — **Astana, Kazakhstan**
B.S. in Mathematical and Comp. Modeling (summa cum laude equivalent) | GPA: 3.86/4.00

AWARDS & HONORS

- Graduate Dean's Research Travel Grant, UC Santa Cruz, USA — 2026
- DAC 2025 Young Fellow, Design Automation Conference, San Francisco, USA — 2025
- IEEE WIE Student Scholarship, International Leadership Conference, San Jose, USA — 2025
- Graduate Studies DEI Research & Travel Award, UC Santa Cruz, USA — 2025
- Divisional MIP Fellowship, UC Santa Cruz, USA ($\tilde{\$}$20K) — 2023
- POSCO Asia Fellowship, South Korea — 2023
- KIST–KT&G Global Scholarship Foundation, South Korea — 2021
- Sur–Place Konrad Adenauer Foundation Scholarship, Germany — 2019
- ITEC Programme Scholarship, Government of India — 2018
- Academic Achievement Award (Seven times), ENU, Kazakhstan — 2018
- Foundation of the First President of Kazakhstan Scholarship — 2018

PUBLICATIONS

(* indicates equal contribution)

1. **[APL Machine Learning 2026] A. Kembay**, S. Gunasekaran, R.-J. Zhu, Y. Zhang, J. K. Eshraghian. (2026). *Efficient Knowledge Distillation via Salient Feature Masking*, APL Machine Learning 4, 016104 (2026). [paper] [code]
2. **[Nature Communications 2025]** S. Gunasekaran, **A. Kembay**, et al. *A Predictive Approach to Enhance Time-Series Forecasting*, Nature Communications, 16(1), 8645. [paper] [code]
3. **[Under Review]** Zhu R.-J.*, ..., **Kembay A.**, ..., Eshraghian J. (2025). *A Survey on Latent Reasoning.* [paper] [code]
4. **[Under Review]** Y. Tian, **A. Kembay**, N. D. Truong, J. K. Eshraghian, O. Kavehei. (2025). *Learning with Spike Synchrony in Spiking Neural Networks.* [paper] [code]
5. **[ISCAS 2025, DAC 2025] A. Kembay***, K. Aguilar*, J. Eshraghian. *A Quantitative Analysis of Catastrophic Forgetting in Quantized Spiking Neural Networks*, 2025 IEEE International Symposium on Circuits and Systems (ISCAS). [paper] [poster] (also presented at DAC Young Fellows Poster Session)
6. **[BayLearn 2024] A. Kembay**, R.-J. Zhu, N. Kuipers, J. Eshraghian, C. Josephson. *Leveraging Spiking Neural Networks for Solar Energy Prediction in Agriculture*, Bay Area Machine Learning Symposium (BayLearn 2024). [paper] [code]
7. **[IWCN 2021]** Sch. Kim, Ch. Lee, B. Lee, D. Seol, D. Kim, **A. Kembay**, K. Yun, S. Jang, J. Lee. *Simulation Web Platform for the Electro-Chemical Oxygen Reduction Reaction*, 2021 International Workshop on Computational Nanotechnology (IWCN), **Oral**. *(Associated patent, 20% share).* [paper] [Project Page] [Platform Showcase]

PATENTS

Electronic Structure Calculation Web-Program, Korea Institute of Science and Technology.
Inventors: Kim Sch., **Kembay A.**, Kim S. (20% share each)
[Project Page] [Platform Showcase]

RESEARCH EXPERIENCE

- **Research Associate, Conscium, London, UK** — Mentor: Dr. Max Ward
  Project: Generative modeling and optimization of mRNA sequences — *Remote | Jan. 2026 – Present*
- **Research Scientist Intern, AI Fund, Mountain View, CA, USA** — Mentor: Dr. Andrew Ng
  Project: AI Systems for Vision-Based Reinforcement Learning in Interactive Environments — *Jan. 2026 – Apr. 2026*
- **Graduate Student Researcher, University of California, Santa Cruz** — Mentor: Prof. Jason Eshraghian
  Project: Knowledge Distillation, Continual Learning and Efficient Language Models — *Oct. 2023 – Present*
- **Research Scientist Intern, Korea University Medicine, Seoul, South Korea** — Mentor: Dr. Il-Joo Cho
  Project: Wireless Brain Chip Optimization and Data Transfer Algorithms — *Apr. 2023 – Sep. 2023*

- **Research Assistant, AI Research Group, Korea Institute of Science and Technology** Mentor: Dr. Suhyun Kim
  Project: Data-Free Knowledge Transfer for Neuromorphic Systems *Sep. 2020 – Mar. 2023*
- **Research Intern, Computational Science Research Center, KIST, Seoul, Korea** Mentor: Dr. Seungchul Kim
  Project: Quantum Dot Simulation Platform Development *Mar. 2020 – Aug. 2020*

TEACHING EXPERIENCE

Teaching Assistant, ECE 173: High-Speed Digital Design **UC Santa Cruz**
Spring 2025 *Class size: 47*

MENTORSHIP

Mentored 4 undergraduate research students to date at UCSC.

Mentored Kazakh/Central Asian students (15+) by raising awareness and providing information about grad school in the US/Europe. Mentees have successfully secured prestigious multi-year awards, such as the Graduate Presidential Fellowship (4yr) and the DAAD Scholarship, and gained admission to fully-funded Ph.D. programs at institutions such as Columbia University.

Mentees

Skye Gunasekaran (Undergraduate@UCSC) – [Nature Communications'25] Dec. 2023 – Aug. 2025
Karina Aguilar (Undergraduate@UCSC) – [ISCAS'25] May. 2024 – Dec. 2024
Mariah Regalado (Undergraduate@UCSC) Jun. 2025 – Oct. 2025
Yuchen Tian (Masters@University of Sydney, Australia) Oct. 2024 – present

PROFESSIONAL MEMBERSHIPS

IEEE Graduate Student Member	Region 6 – Western USA, Monterey Bay Subsection
IEEE Women in Engineering (WIE)	Active Member
IEEE Circuits and Systems Society (CASS)	Active Member
Graduate Womxn in Computing, UCSC	Member – Supporting community-building, peer mentorship, and networking for gender minorities in computing-related research

PROFESSIONAL SERVICE

Reviewer: NeurIPS NeuroAI '24, APL Machine Learning '24–25, IEEE ISCAS '24–26, IEEE TCDS '25, ACM TOMM '25